

Python Backend Assignment

Topic: Build a RESTful API with FastAPI/Flask & SQLAlchemy

1. Environment & Setup

Initialize a clean Python environment to ensure reproducibility.

- Create a virtual environment: `python -m venv venv`.
- Manage dependencies using a `requirements.txt` file.
- Use **pydantic** (if using FastAPI) for data validation and settings.

2. Data Persistence (SQLAlchemy)

Model a simplified "Inventory Management System".

- **Item Model:** `id`, `name`, `description`, `price`, `stock_count`.
- **Category Model:** `id`, `name` (Relationship: One Category to Many Items).
- Implement database sessions using a dependency injection pattern.

3. API Endpoints

- `POST /items/`: Create a new item with validation for positive price and stock.
- `GET /items/{id}`: Fetch details of a specific item; return 404 if not found.
- `PATCH /items/{id}/reduce-stock`: Atomically decrease stock when a "sale" occurs.
- `GET /categories/`: List all categories with the count of items in each.

4. Pythonic Standards

- **Type Hinting:** Use Python type hints throughout the project for clarity.

- **Asynchronous Programming:** Use `async / await` for database calls if supported by the driver.
- **Testing:** Include at least three functional tests using `pytest` and `httpx`.